Computer Networks

Lecture-1

Course Outline

- The course will consist of :
  - 16 lectures
  - Atleast 2 quizzes and 1 Presentation
  - 2 exams (1 midterm and 1 final)

- Grading Criteria:
  - Midterm:30%
  - Final Exam: 50%
  - Assignments: 20%

Course Contents

Intro to Networks components, Basic concepts of networking.
Network topologies, network layers, internet standards, and architecture.

The concept of layered architecture modeling. TCP/IP protocol suite, comparison, protocols and OSI.

The distinction between Data and Signals of Analog and Digital Transmission Impairments. Network performance measures including throughput, and delays. Data vs. signaling rates.
Channel bandwidth and capacity, delay product.

Switching Technologies: Circuit Switching, Packet Switching, datagram, and Virtual Circuit Switching: Phases, efficiency, and delays.

Data link layer (DLL), Nodes and links, services, framing, flow,
error, congestion control, data link addressing, ARP, and communication using DLL.

Error Detection and Correction, Types of Error, Redundancy, Error Detection, Parity, checksum, VRC, LRC, cyclic codes, Detection versus Correction, Forward Error Correction
(Hamming codes, Hamming distance) versus Retransmission

Data link control services, Flow, and error control (simple
protocol & stop N wait protocol), PAP and CHAP Protocols, Go back N (GBN) and Selective Reject.

Multiple Control Access: Random access protocols (Aloha, CSMA, CSMA/CD, CA), Wireless hidden and exposed node problems, and Controlled access protocols (Reservation, Polling, and Token Passing)

Ethernet, Evolution, Standard Ethernet addressing, access method
Fast Ethernet Access, Mac sub-layer.

Wlans: architecture, characteristics, 802.11 sublayers, Exposed and hidden node problem.v

Connecting devices: Hub, Switch and Router, Virtual Lans, Spanning tree Algorithm.
IP addressing, classful, classless routing, local and global addresses including APIPA.

IP addressing format, fragmentation, ICMP v4: error reporting and query messages.

Routing: Least cost, Routing Algorithms: Distance vector and link state, RIP and RIP messages, algorithm, and timers

Textbook

- Computer Networking: A Top-Down Approach Featuring the Internet, 7th Edition by James F. Kurose and Keith W. Ross
- Computer Networks, 5th Edition by Andrew S. Tanenbaum
- Data and Computer Communications, 10th Edition by William Stallings
- Data Communication and Computer Networks, 5th Edition by Behrouz A. Forouzan

Overview of Lecture 1

- Data Communication
- Data Communication System
- Factors effecting Data Communication system
- Components of data communication system
- Data Representation

COMMUNICATION

Sharing of information

(local or remote)

Telecommunication

Communication at a distance
(includes telephony, telegraph, and television etc.)

DATA COMMUNICATION

"The exchange of Data (e.g. in form of 0's and 1's) from one entity (e.g. Computer) to the other using a Transmission Medium"

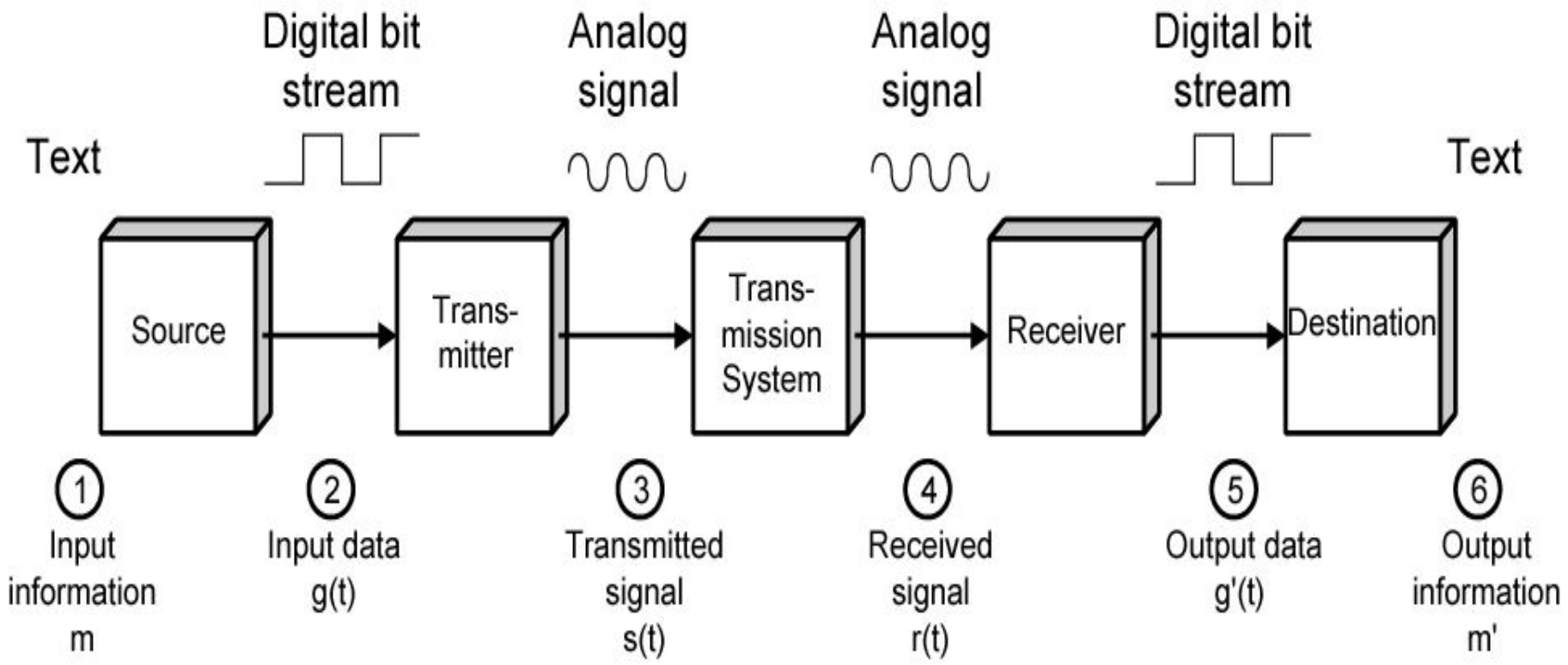
Data Communication (cont..)

- “Simply means sharing of data locally or remotely.”
- LOCAL
  - communication usually occurs face to face
  - Communicating devices are present in the same building or a similarly restricted geographical area.
- REMOTE
  - remote communication takes place over distance.
  - Communicating devices are present farther apart.
  - The term telecommunication, which includes telephony, telegraphy, and television, means communication at a distance (tele is Greek for "far").

Data Communication System

For Data Communication to occur, communicating devices must be a part of a communication system made up of a combination of hardware (physical equipment) and software (programs). This system is known as "DATA COMMUNICATION SYSTEM"

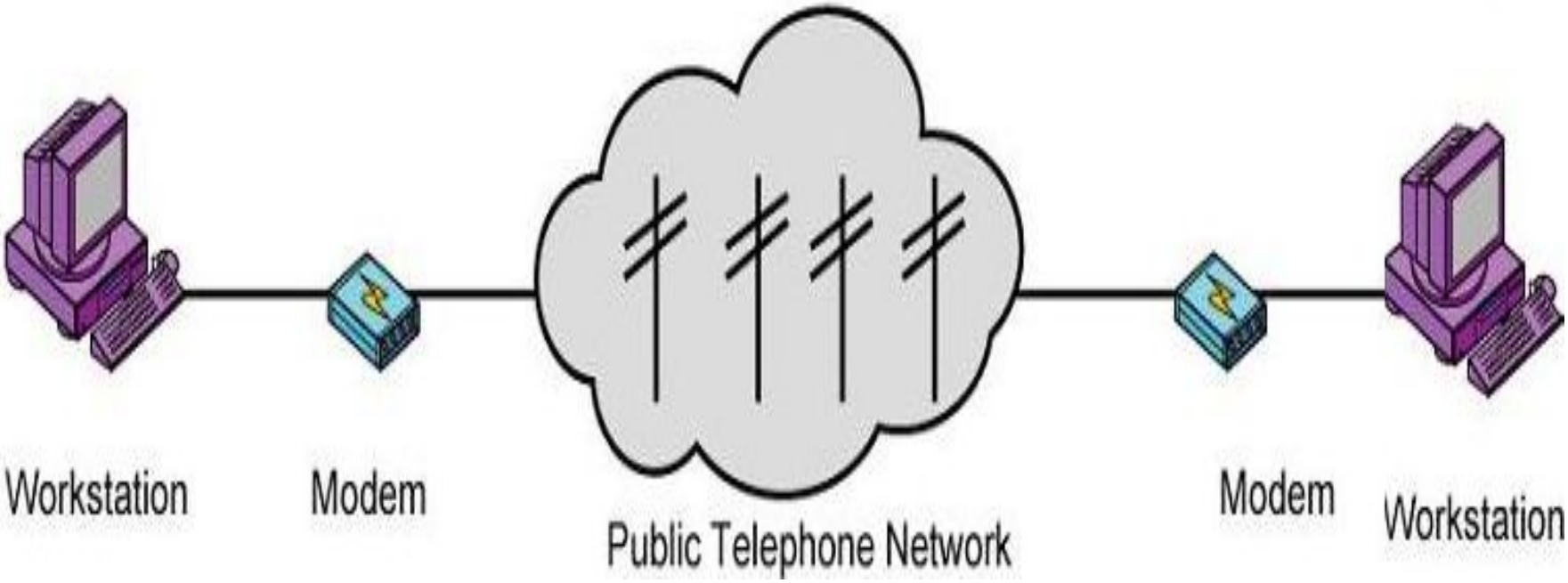
Data Communication System



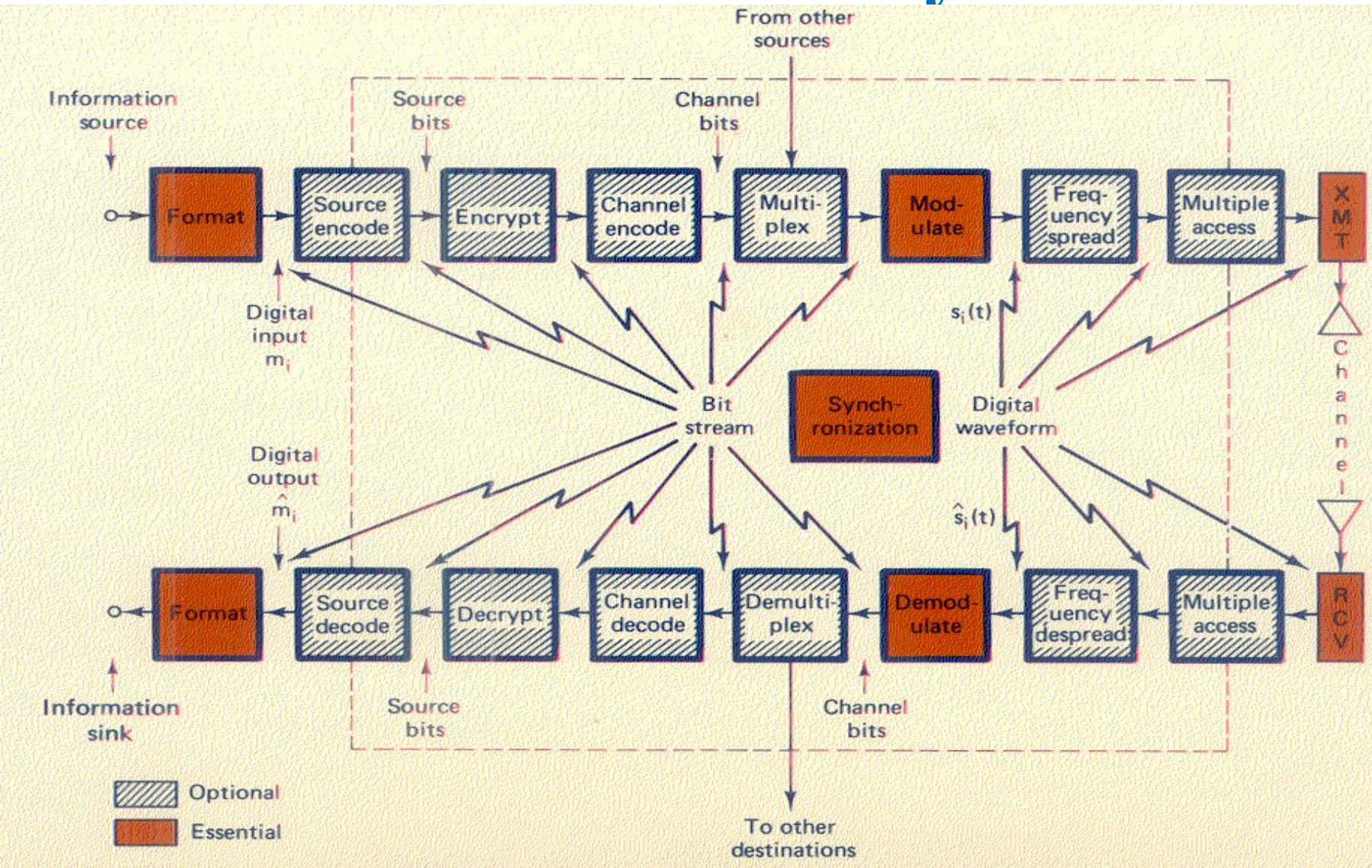
Components of a Data Communication System

- Source
  - generates data to be transmitted
- Transmitter
  - Converts data into transmittable signals
- Transmission System
  - Carries data
- Receiver
  - Converts received signal into data
- Destination
  - Takes incoming data

EXAMPLE – Electronic Mail



An Actual Digital Data Communication System



- **Information Source**
  - The origin of the message or data to be transmitted.
- **Format (Essential)**
  - Converts the original message (e.g., audio, text, video) into a digital format suitable for processing.
- **Source Encode (Optional)**
  - Compresses data to remove redundancy (e.g., Huffman coding or JPEG).
- **Encrypt (Optional)**
  - Secures the data for confidentiality using encryption techniques.
- **Channel Encode (Optional)**
  - Adds redundancy to detect/correct errors introduced by the channel (e.g., Hamming, Turbo codes).

- **Multiplex (Optional)**
  - Combines multiple data streams into one for efficient transmission.
- **Modulate (Essential)**
  - Converts digital bits into analog signals (waveforms) to match the physical channel (e.g., PSK, QAM).
- **Frequency Spread (Optional)**
  - Spreads signal across a wide frequency band to resist interference (e.g., Spread Spectrum).
- **Multiple Access (Optional)**
  - Shares channel resources among multiple users (e.g., TDMA, FDMA, CDMA).
- **Transmit (XMT) (Essential)**
- Sends the signal into the communication channel.

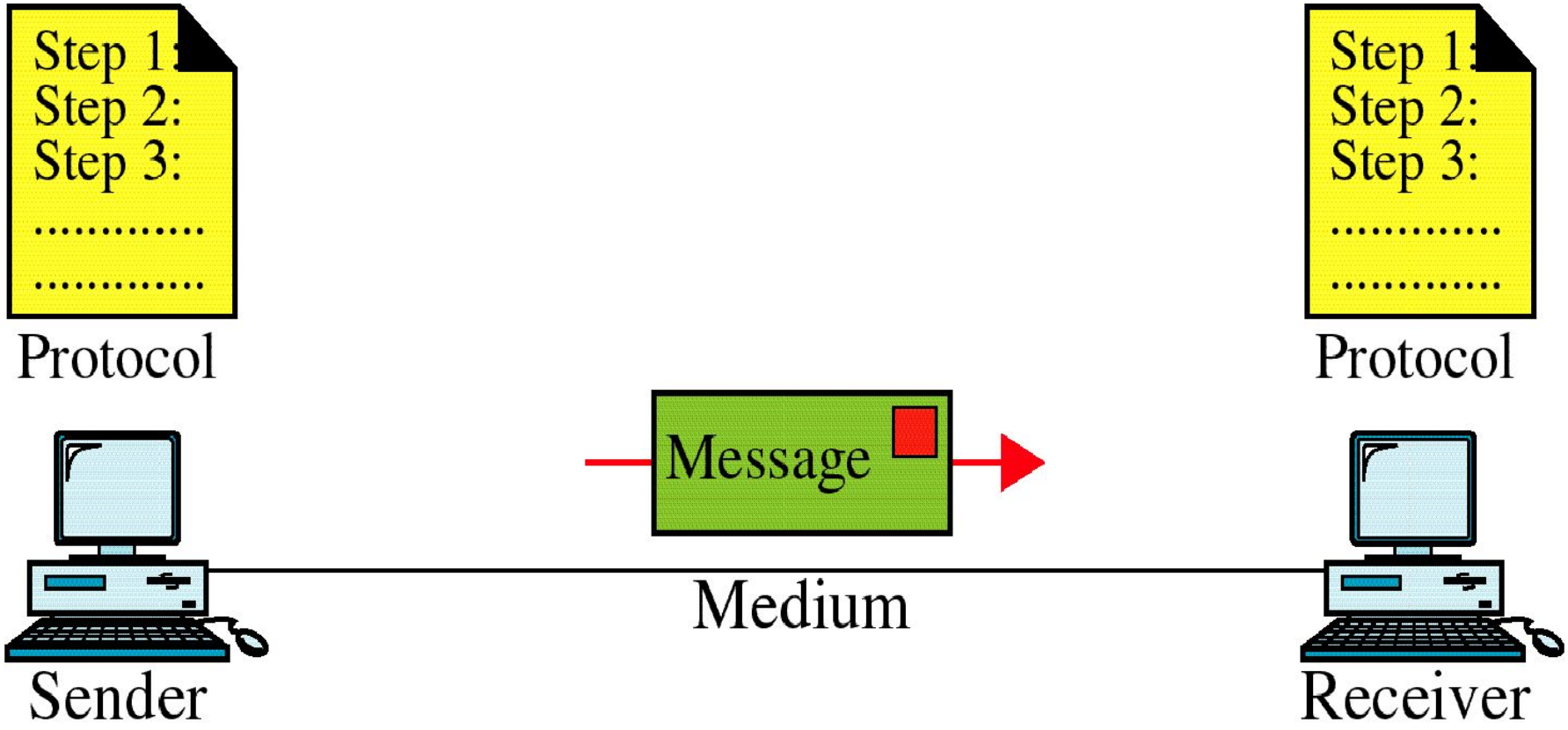
Effectiveness of Data Comm. System

- Effectiveness depends upon four fundamental characteristics:

1. Delivery (to the specific/ correct destination)
2. Accuracy (correctly/ accurately)
3. Timeliness (data delivered in a timely manner, without significant delay)
4. Jitter (uneven/variation delay in the delivery of audio or video packets)

- For example a postal mail.

Components of a Simple Data Communication System



Components of a Data Communication System

- A Data communication system is made up of 5 components:

1. Message
2. Sender
3. Receiver
4. Medium
5. Protocol

Message

- The message is the information (data) to be communicated. A message is transmitted from sender to receiver.
- It can be
  - data file
  - an audio or video file
  - or any text message

"A message also keeps information like sender and destination addresses."

Sender

- It is also called “source”.
- It is the device which generates the message and responsible for its transmission.
- It can be a
  - computer
  - Workstation
  - telephone handset
  - video camera
  - or any communication device.

Receiver

- It is also called “destination/sink”.
- It is the device that receives the transmitted message.
- It may be a
  - computer
  - Workstation
  - telephone handset
  - Television
  - or any communication device.

Medium

- The transmission medium is the physical path by which a message travels from sender to receiver. It may be a
- guided medium
  - coaxial cable
  - twisted pair cable
  - fiber optic
- unguided medium
  - electro-magnetic waves
  - micro waves
  - infrared waves

Protocol

- Protocols are also called language of communication.
- A protocol is a set of rules that governs data communication.
- Both sender & receiver must have the same protocol for communication.
- It represents an agreement between the communicating devices.
- Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Data Representation

Data Representation

- Information today comes in different forms such as;

1. Text
2. Numbers
3. Images
4. Audio
5. Video.

Text

- In data communications, text is represented as a bit pattern, a sequence of bits (O's or I's).
- Different sets of bit patterns have been designed to represent text symbols. Each set is called a code, and the process of representing symbols is called coding.
- Today, the prevalent coding system is called Unicode, which uses 32 bits to represent a symbol or character used in any language in the world.
- The American Standard Code for Information Interchange (ASCII), constitutes the first 127 characters in Unicode and is also referred to as Basic Latin.

Class Activity

- Explore Unicode, ASCII Code

Numbers

- Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers;
- The number is directly converted to a binary number to simplify mathematical operations.

Images

- Represented by bit patterns.
- Composed of a matrix of pixels (picture elements).
- Pixel is a small dot.
- The size of the pixel depends on the resolution. For example, an image can be divided into 1000 pixels or 10,000 pixels.
- Better representation of the image (better resolution), more memory is needed to store the image.

- After an image is divided into pixels, each pixel is assigned a bit pattern.
- The size and the value of the pattern depend on the image.

- For an image made of only black and white dots (e.g., a chessboard), a I-bit pattern is enough to represent a pixel.

Images

- If an image is not made of pure white and pure black pixels, you can increase the size of the bit pattern to include gray scale. For example, to show four levels of gray scale, you can use 2-bit patterns.
- A black pixel can be represented by 00, a dark gray pixel by 01, a light gray pixel by 10, and a white pixel by 11.

- There are several methods to represent color images. One method is called RGB, so called because each color is made of a combination of three primary colors: red, green, and blue.

- The intensity of each color is measured, and a bit pattern is assigned to it. Another method is called YCM, in which a color is made of a combination of three other primary colors: yellow, cyan, and magenta.

Audio

- Audio refers to the recording or broadcasting of sound or music.
- Audio is by nature different from text, numbers, or images. It is continuous, not discrete.
- Even when we use a microphone to change voice or music to an electric signal, we create a continuous signal.

Video

- Video refers to the recording or broadcasting of a picture or movie.
- Video can either be produced as a continuous entity (e.g., by a TV camera),
- or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.

Summary

- Data Communication
- Data Communication System
- Factors effecting Data Communication system
- Components of data communication system
- Data Representation

Suggested Reading

- Section
  - 1.1,
    - “Data Communications and Networking” 5th Edition by Behrouz A. Forouzan

- Sections
  - 1.1,
  - 1.2,
    - “Data and Computer Communication” 6th Edition by William Stallings